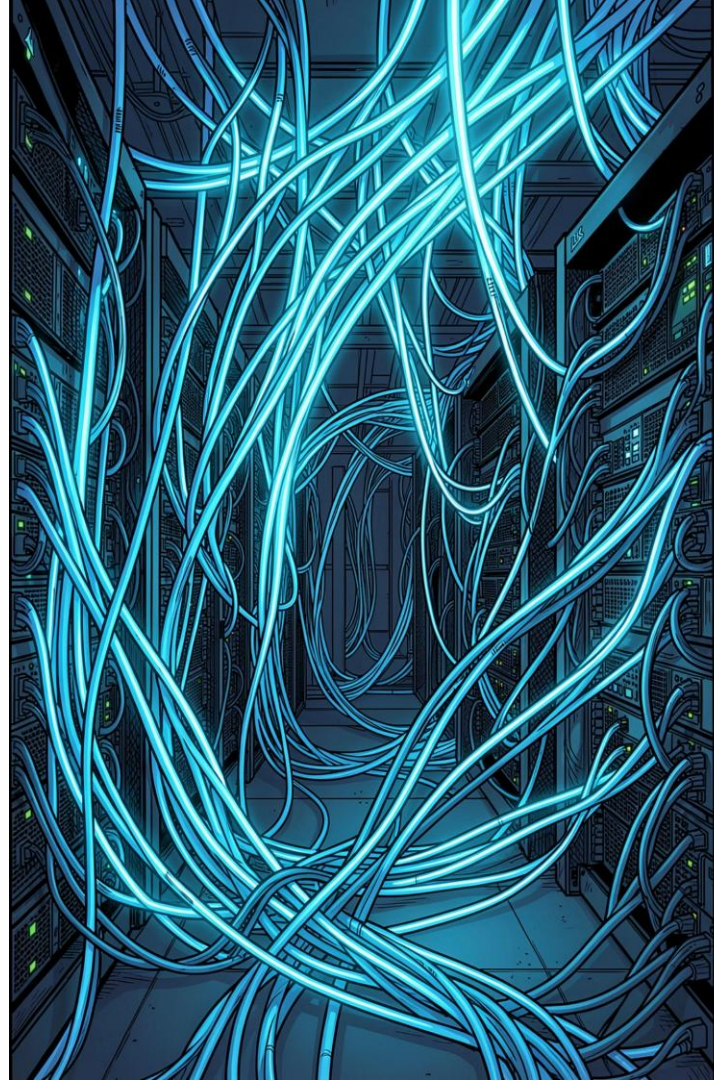


BUSINESS ANALYTICS PORTFOLIO PROJECT

Customer Churn Analytics Dashboard

Identifying High-Risk Customers, Revenue at Risk, and Retention Opportunities Through SQL, Excel, and Power BI

Prepared by **Shivam Yadav** · Business Analytics Portfolio Project



Project Overview & Key Outcomes

This project addresses customer churn in a mid-sized telecom operator through end-to-end analytics — from data cleaning and SQL querying to Power BI visualization. The goal: identify churn drivers, quantify financial exposure, and recommend retention strategies.

7,043

Customers Analyzed

Full dataset spanning demographics, contracts, services, and payment methods

26.54%

High-Risk Rate

Proportion of customers identified as high churn probability

₹139.1K

Revenue at Risk

Monthly recurring revenue exposed to attrition

1,869

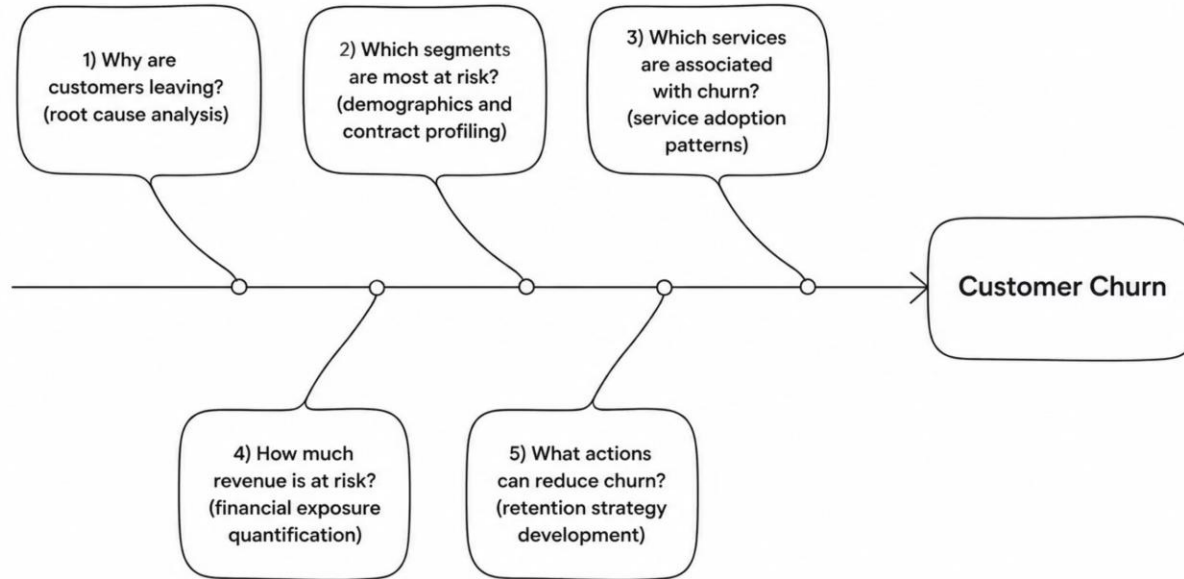
High-Risk Customers

Individual accounts flagged for proactive retention intervention

Primary Insight: Month-to-Month contract customers represent the highest-risk segment, with churn rates nearly double the overall average — a clear intervention priority.

The Churn Challenge: Strategic Questions

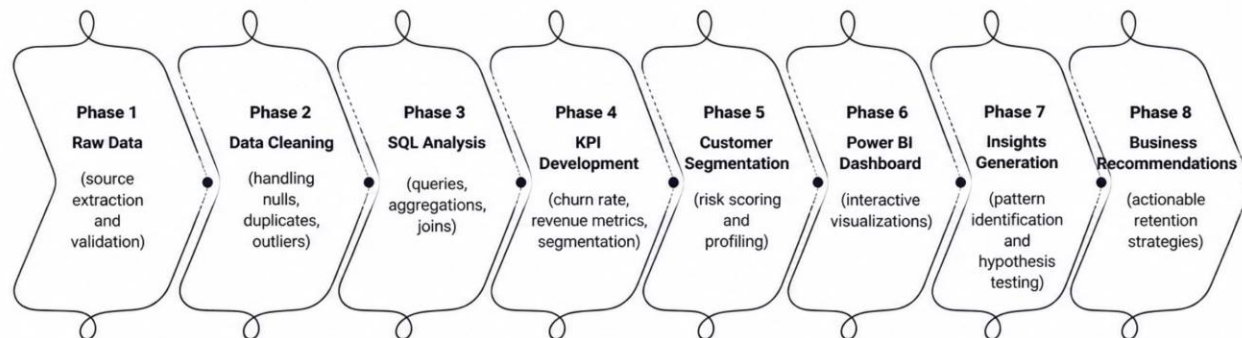
The telecom operator faced accelerating customer attrition, eroding customer lifetime value and compressing margins. This project was structured around five strategic questions to guide the analytical framework.



Answering these questions required a structured analytical approach combining SQL for data extraction, Excel for exploratory analysis, and Power BI for interactive executive dashboards.

Project Approach: End-to-End Analytics Framework

This project followed a structured eight-phase methodology – from raw data ingestion through actionable business recommendations – ensuring analytical rigor and executive-ready outputs at every stage.



Each phase produced documented deliverables, enabling reproducibility and stakeholder review. The final Power BI dashboard serves as a living analytical asset for ongoing churn monitoring.

Dataset Profile: 7,043 Customer Records

The dataset spans seven analytical dimensions, providing a comprehensive view of customer behavior, contract structure, and service adoption patterns across the telecom subscriber base.

Demographics

Age, gender, senior citizen status, and partner/dependent indicators for population profiling

Contract Data

Contract type (Month-to-Month, 1-Year, 2-Year), tenure in months, and paperless billing status

Service Usage

Phone, internet, streaming, security, backup, and device protection service subscriptions

Payment Methods

Electronic check, bank transfer, credit card, and auto-pay enrollment status

Revenue Metrics

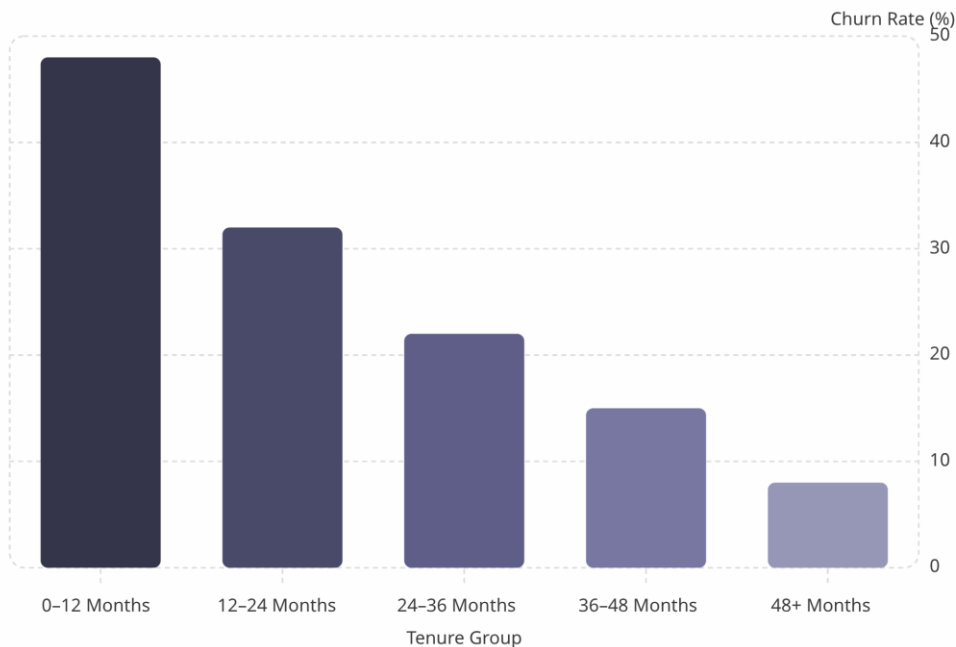
Monthly charges, total charges, and tenure-adjusted revenue per customer segment

Churn Status

Binary churn flag enabling supervised segmentation and risk modeling

KEY FINDING #1

Customer Tenure Is the Strongest Churn Predictor



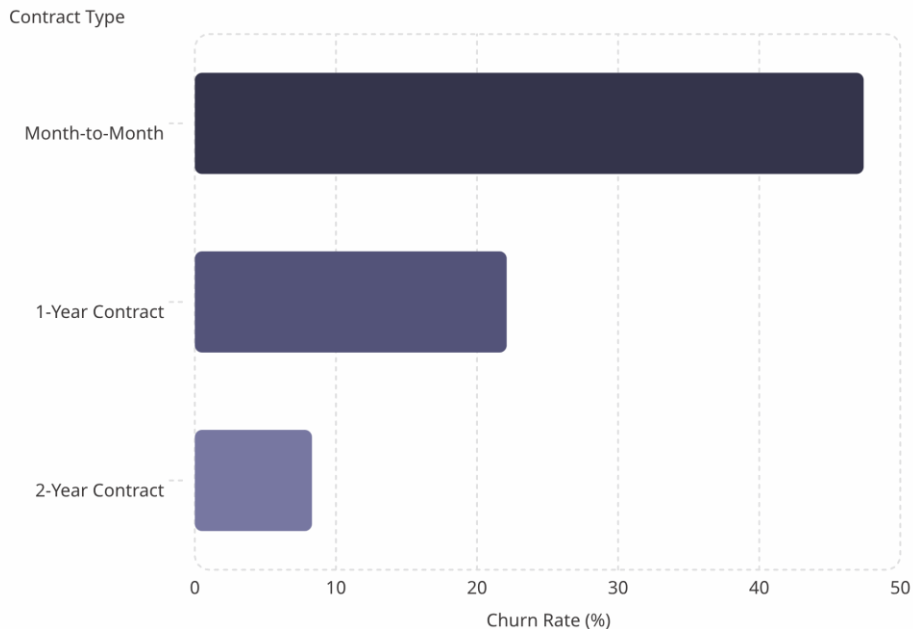
Strategic Insight

Customers in their first 12 months carry a churn rate of **~48%** – six times higher than the 48+ month cohort. Churn declines monotonically with tenure, confirming the critical importance of early-stage retention.

- First-year customers require proactive onboarding and engagement programs
- Churn intervention ROI is highest within the first 90 days
- Long-tenure customers represent the most stable, high-LTV segment

KEY FINDING #2

Contract Type Is the Primary Structural Churn Driver



47.4% Churn Rate

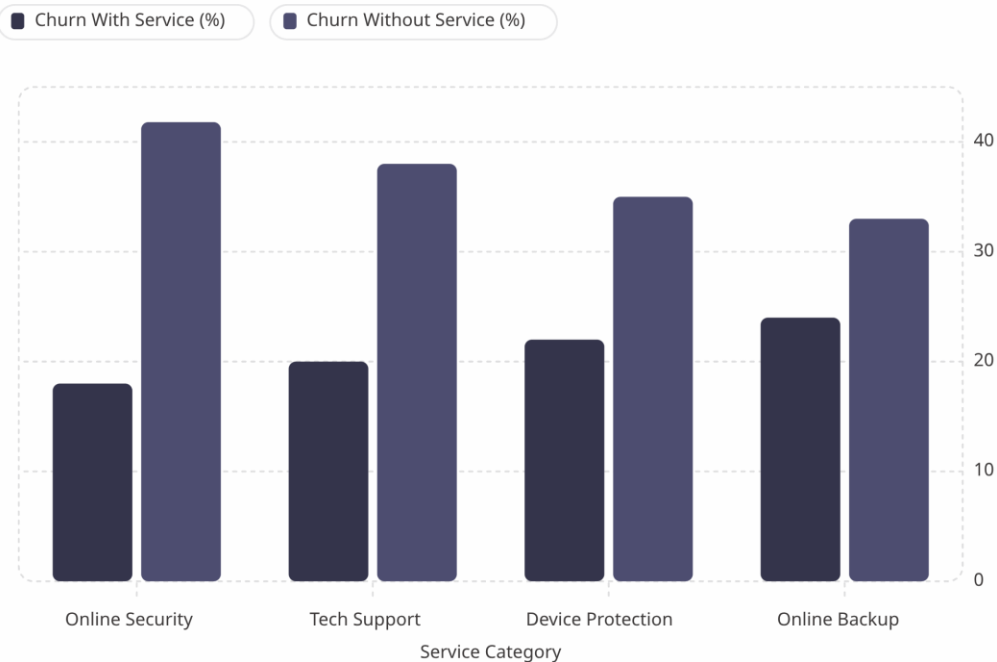
Month-to-Month customers churn at nearly **six times** the rate of 2-Year contract holders — making contract structure the single most impactful retention lever.

Long-term contracts create structural switching costs that dramatically reduce voluntary attrition, even among otherwise dissatisfied customers.

📌 **Strategic Implication:** Converting Month-to-Month customers to annual contracts is the highest-ROI retention action available.

KEY FINDING #3

Service Adoption Drives Customer Stickiness



41.8% Churn Without Online Security

Customers lacking Online Security churn at more than **double** the rate of those enrolled – the strongest service-level churn signal in the dataset.

Bundled service adoption creates dependency and perceived value, raising switching costs and reducing price sensitivity.

- Online Security: highest individual impact on retention
- Tech Support and Device Protection reinforce stickiness
- Service bundles outperform standalone promotions



FINANCIAL IMPACT

Revenue at Risk: ₹139.1K Monthly Exposure

The 1,869 high-risk customers identified represent a measurable and addressable monthly revenue exposure. Reducing churn by even 20% would recover approximately **₹27.8K per month** – or over **₹333K annually** – without acquiring a single new customer.

₹139.1K

Monthly Revenue at Risk

Recurring revenue exposed across all high-risk customer segments

1,869

High-Risk Accounts

Customers flagged for proactive retention outreach and intervention

26.54%

Overall Churn Rate

Baseline attrition rate across the full analyzed customer base

₹333K

Annual Exposure

Projected yearly revenue loss if current churn trends continue unchanged

Five Prioritized Retention Actions

1

Retention Offers for Month-to-Month Customers

Deploy targeted discounts and loyalty incentives to convert high-risk Month-to-Month subscribers to annual contracts, directly addressing the 47.4% churn segment.

2

Strengthen First-Year Onboarding

Implement structured 90-day engagement programs — welcome sequences, usage tutorials, and proactive support — to reduce the 48% first-year churn rate.

3

Promote Online Security & Service Bundles

Cross-sell Online Security, Tech Support, and Device Protection during onboarding and renewal windows to increase switching costs and reduce the 41.8% service-level churn rate.

4

Drive Auto-Pay Migration

Incentivize electronic payment enrollment to reduce involuntary churn from payment failures and improve payment consistency across at-risk segments.

5

Personalized Retention Campaigns

Leverage the Power BI dashboard for segment-level churn scoring and deploy personalized outreach — email, SMS, and call-center triggers — based on real-time risk profiles.

"Turning Customer Data into Retention Strategy" — Data-driven retention actions can reduce churn, protect ₹139.1K in monthly revenue, and improve long-term customer lifetime value.